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1. Introduction

1.1 Background

The ability to classify images into pre-specified classes using their visual content is a crucial aspect of image classification task of computer vision. It is of extreme importance to many applications such as Autonomous navigation, traffic monitoring, face detection, and medical image analysis. The traditional image classification method is based on high resolution images and large memory which needs computational resources and memory. However, compressed domain image classification is a better approach as the capabilities of various processors and components evolve. As technologies mature and different processors and components become available, it becomes more practical to use storage efficiently and leverage processing capabilities.

In general, images are transmitted or stored as compressed versions in standard apps so as to minimize their file sizes. These are typically decompressed prior to being processed in the case of using deep learning models for image classification, which adds to the computing load. The goal of this research is to determine whether compressed images can be classified directly without complete decompression, and whether image classification can be performed in the compressed domain without compromising on classification accuracy using a Deep Separable Convolutional Neural Network (DS-CNN).

1.2 Motivation

The growing demand for efficient image processing methods, especially for low-light conditions such as power and resource-limited systems, was the inspiration for this work. Image classification is needed for Edge computing systems, smartphones with limited resources and cloud services. Thus, the efficacy of image-based applications across many industry sectors would be greatly enhanced through high compression-domain classification accuracy.

Within the medical imaging field, DR is a chronic eye disease that attacks the retina and is a long-term complication of diabetes which leads to Loss of sight and blindness and this must be prevented by detecting it at an early stage. The large retinal scans are to be sent from remote clinics to central servers where it can be analyzed by AI which is essential for telemedicine applications. This study investigates whether the reduced visual quality of SVD-compressed images negatively impacts the diagnostic accuracy of deep learning models.

1.3 Problem Statement

This study aims to explore the dataset APTOS 2019 Blindness Detection dataset from Kaggle to investigate the effect of compression on classification. The test of the DS-CNN model is conducted and trained in order to see how well it performs in recognizing and classifying a series of retinal Images compressed with various qualities using SVD-based sparse compression sampling. To conserve disk space and retain important image characteristics, the image was cropped. The image was cropped to capture important image features while minimizing disk space and file size. The research tries to balance the levels of classification accuracy and compression.

1.4 Objectives

The main aims of this research are:

- To design and implement an architecture of custom DS-CNN with Squeeze-and-Excitation(SE) blocks which are used to classify DR.
- To apply an SVD-based sparse sampling image compressing algorithm that achieves.
- High compressions with minimum loss of visual elements. To compare the classification results of different levels of compression.
- The goal of this experiment is to show that deep learning models can be used to classify compressed images successfully without needing complete decompression.
- To determine the best compression ratio for a good weight loss, size reduction and classification efficiencies.

1.5 Scope of the Project

This research is our effort to establish a relationship between accurate classification of images by using a trained neural network and storing images in compressed form to reduce the amount of computational resources needed, by classifying the compressed images directly. Our research in this domain may result in newer applications in other fields in which scarcity of resources is present and efficient utilisation is important, such as IoT systems, transportation networks applications, real-time monitoring and medical imaging/diagnosis.

1.6 Organization of the Report

- **Chapter 2** presents a comprehensive literature review of existing work in compressed domain image classification, SVD-based compression, and deep learning architectures.
- **Chapter 3** details the methodology, including dataset preparation, compression algorithm, model architecture, and training procedure.

- **Chapter 4** presents the experimental results and discussion.
- **Chapter 5** concludes the report with key findings and future scope.

2. Literature Review

2.1 Genetic Algorithm-Based Sensing Mask Learning

This study we reviewed was on the classification of images from sensing measurements without the need for the images to be reconstructed. It appeared in Applied Sciences journals. The solution provided improved the classification accuracy by using binary masking optimization using a genetic algorithm, which reduced the amount of resourced needed for processing. The genetic algorithm approach incrementally improve the sensing mask so that important features are retained, improves the efficiency and accuracy. It is an appropriate for the application of resource-limited systems such as edge computing, mobile phones, IoT devices and so on.

2.2 A Comprehensive Survey of Convolutions in Deep Learning

The survey paper (2024) [5] is a detailed study of the convolutional operations that are the foundation of today's Computer Vision field. It describes the growth and development of Convolutional Networks (CNNs) from the classical 2D CNNs to specialized models for specific problems such as multi-scale feature extraction. For certain medical applications, they discovered that lightweight and efficient architectures with advanced variants of convolution perform better than basic CNNs, but at the cost of loss in diagnostic performance. This directly inspired us to pick Depthwise Separable Convolutions for classification.

2.3 Medical Image Compression Using SVD

The motivation of our research is clearly stated in the study on medical image compression by Singular Value Decomposition (2023) [6]: There is a need to compress the high-resolution medical images to facilitate efficient telemedicine. When applied to DR screening, the large retinal scans need to be sent from remote clinics to central servers for AI analysis. The authors prove that SVD is a suitable mathematical tool for the task with the added advantage of tunable storage-to-image fidelity tradeoff. The next important question that we must address will be: "Does the lowered image quality of SVD compressed images affect the diagnostic performance of Deep Learning models?" based on this work.

2.4 Squeeze-and-Excitation Networks

At IEEE CVPR, Hu, Shen, and Sun [11] (2018) proposed the Squeeze-and-Excitation Networks that dynamically adjust the response of each channel based on the interdependency between channels. The SE block is followed by a "squeeze" operation which aggregates spatial information

through global average pooling and an "excitation" operation which learns channel-wise attention weights through two fully connected layers. It lets the network focus on informative features and ignore less informative ones, thus reducing the computational cost while boosting the classification accuracy.

2.5 Deep Learning for Diabetic Retinopathy Detection

This study is about deep learning and its applications in the field of diabetic retinopathy detection. Several recent works [12] [13] [14] have delved into deep learning-based automated detection and grading of DR. These findings show that the CNN can outperform experts in DR severity classification using fundus images. But most of the current methods rely on high-resolution images, thus limiting the use in low-resource telemedicine environments. This is where our work comes in, showing that the classification of images could be preserved even in the case of their significantly compressed images.

2.6 Research Gap

Although existing literature proves the feasibility of compressed domain classification and the effectiveness of deep learning in detecting diabetic retinopathy, few studies have integrated SVD-based sparse sampling compression with lightweight DS-CNN model with SE attention mechanism for medical image classification. To fill this gap, we show that a custom DS-CNN model with SE blocks can achieve high classification accuracy on SVD-compressed retinal images for a wide range of compression levels.

3. Methodology

3.1 Overview

The research discusses the effect of image compression on classification performance with the Deep Separable Convolutional Neural Network (DS-CNN) model. The methodology includes the model architecture, training, evaluation metrics, image compression, and dataset preparation.

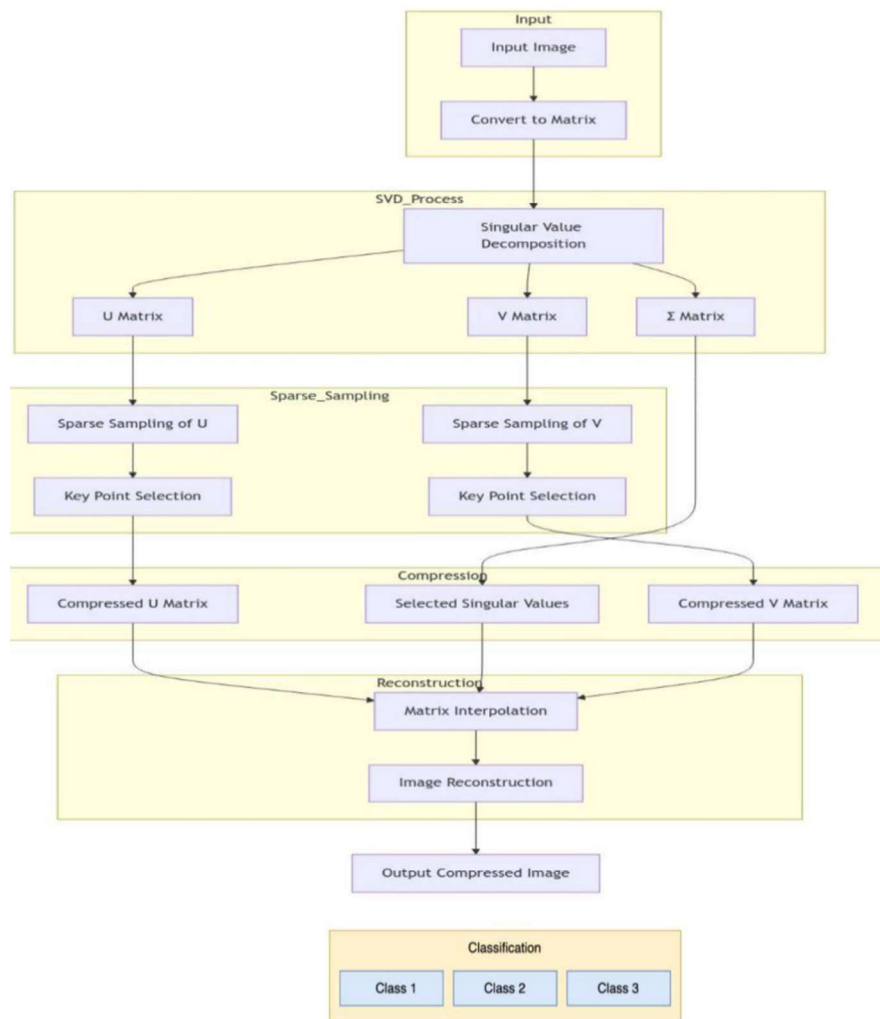


Fig. 3.1: Complete architecture overview

3.2 Dataset Preparation

The dataset is obtained from the APTOS 2019 Blindness Detection competition on Kaggle, which aimed to encourage the development of machine learning-based automated diabetic retinopathy

detection from retinal images. A Depthwise Separable Convolutional Neural Network (DS-CNN) is trained on the dataset to make predictions about the stages of diabetic retinopathy. The retinal color fundus (back of the eye) images used are stored in the dataset's "coloured_images" folder.

Dataset Specifications:

- **Source:** APTOS 2019 Blindness Detection (Kaggle)
- **Image Type:** Color retinal fundus images
- **Image Resolution:** Resized to 224×224 pixels
- **Number of Classes:** 5 (No DR, Mild, Moderate, Severe, Proliferative DR)
- **Total Images:** 3,662 (2,931 training + 731 validation)
- **Split Ratio:** 80% training, 20% validation (using validation_split=0.2)

Preprocessing:

The images were preprocessed and resized to 224×224 pixels to make them usable for training deep learning models. The following preprocessing steps were applied:

1. **Rescaling:** Pixel values normalized to [0, 1] range by dividing by 255.

2. Data Augmentation (Training only):

- Rotation range: $\pm 20^\circ$
- Horizontal flipping
- Zoom range: 0.2
- Brightness range: [0.8, 1.2]

3. **Validation data:** Only rescaling applied (no augmentation) to ensure objective evaluation.

The data augmentation is used to artificially enhance diversity since medical datasets are normally smaller in size. Rotation helps the model function despite any extra rotations in the sample images, zooming/brightness adjustments simulate real world variations in image capture conditions such as dimly lit areas in villages or bad camera quality.

3.3 Image Compression

Images are compressed using an SVD-based sparse sampling technique at various levels of compression to examine classification performance in the compressed domain.

3.3.1 Sparse Sampling of Singular Vectors

The compression technique implements sparse sampling of singular vectors to produce higher quality output of compressed images, retaining more of the image's structural content and achieving higher compression ratios without losing too much image quality.

Singular vectors are processed as sparse representable signals. This method does not remove singular vectors that have small singular values, rather it selects the most relevant points in the left (U) and right (V) singular vectors.

Sparse selection of Key Points:

The key points in every column of singular vector matrix are selected based on their signal characteristics. The algorithm selects key points that have a significant variation in the signal: it contains a condition that identifies turning points in the signal, as the product of adjacent differences changes sign.

Sparse Representation of Left and Right Singular Vectors:

Here each column of the singular vector matrices U and V is sparsely represented, while preserving the structure of the matrix. Only the key points are stored in the sparse matrices U_{sparse} and V'_{sparse} .

3.3.2 Reconstruction via Interpolation

Upon decompression, the compression technique rebuilds the image using linear interpolation along the sparse singular vectors. The interpolation algorithm calculates missing entries in the matrices based on the key points obtained during the sparse sampling process.

Direction and Increment Calculation:

To calculate direction and increment between important points, the direction is computed by examining the sign of the difference of consecutive important points, and increment as the average of the differences between them.

Interpolation of Sparse Matrices:

Using the direction and increment, missing values within U' and V' are interpolated to approximate the full singular vector matrices.

Final Reconstruction:

After reconstructing U' and V' , the compressed image is reconstructed as:

$$A_{\text{compressed}} = U' \times \Sigma \times V'^T$$

3.3.3 Compression Ratio Calculation

The compression ratio ρ is calculated based on the number of key points sampled. For a sparse representation of matrices with sampling frequency p , the formula becomes:

$$\rho = (m \times n) / ((m \times p_m + n \times p_n + 1) \times r)$$

where p_m and p_n are the proportions of the key points in U' and V' respectively, and r is the number of retained singular values. This enhanced compression ratio is achieved by only storing sparse singular vectors rather than full matrices.

3.3.4 Compression Code Implementation

The proposed algorithm is:

Compression Stage:

1. Decomposition: Perform SVD on the original image matrix to obtain U , Σ , V^T
2. Sparse Sampling: Extract key points from the singular vectors to form U'_{sparse} and V'_{sparse}
3. Storage: Store U'_{sparse} , Σ , and V'_{sparse} as the compressed representation

Decompression Stage:

1. Interpolation: Reconstruct U and V from sparse representations.
2. Image Reconstruction: Recreate the compressed image using matrix multiplication of the reconstructed U , Σ , and V^T .

3.4 Model Architecture

The classification model is built upon the Depthwise Separable Convolutional Neural Network (DS-CNN) architecture, enhanced with Squeeze-and-Excitation (SE) blocks to improve feature representation.

3.4.1 Depthwise Separable Convolutional Neural Network (DS-CNN)

The conventional CNN is expensive to train, due to its application of a full convolution for each channel of the input image. In other words, Depthwise Separable Convolutions (DS-CNNs) achieve the efficiency of optimizations by separating depth-wise processing from spatial feature extraction.

The DS-CNN is composed of:

- **Depthwise Convolution:** Performs at least half the computation, applies one filter to each of the input channels (3×3 kernel).
- **Pointwise Convolution:** Multiplies the output of depth-wise convolutions with 1×1 convolutions to produce feature maps.
- **Batch Normalization:** Normalize Activations for Stabilizing Learning.
- **ReLU Activation:** Adds non-linearity to enhance learning.

The stacking multiple DS-CNN blocks leads to the model maintaining high accuracy while reducing the computational complexity. It can be particularly helpful for embedded and mobile AI applications where computational power is low.

3.4.2 Squeeze-and-Excitation (SE) Blocks

SE blocks dynamically compute channel-wise features, which improves the importance of features learnt. The SE mechanism consists of:

1. **Global Average Pooling (GAP):** Importance aggregated by computing a scalar for each channel.
2. **Fully Connected Layers:** Support interdependency among channels by including non-linearly transformable components (reduction ratio = 16).
3. **Sigmoid Activation:** Generates attention weights for each channel.

4. **Multiplication with Input features:** Rescales feature maps which highlight the most informative features.

3.4.3 Class Imbalances

Class imbalance is a typical case in medical data, where some classes (e.g., "No DR") occur more frequently than others (e.g., "Proliferative DR"). Class weights are computed using scikit-learn's `compute_class_weight` function with the 'balanced' strategy to prevent the model from leaning towards majority classes. The formula assigns higher weights to classes with lower samples, ensuring proper training with good accuracy.

3.5 Training Procedure

The following procedure is employed for training the DS-CNN model:

Optimizer: Adam optimizer with initial learning rate of 0.001, which dynamically adapts the learning rate during training.

Loss Function: Categorical cross-entropy loss for multi-class classification.

Metrics:

- AUC (Area Under the Curve) - used as the primary accuracy metric
- F1-Score
- Precision
- Recall

Callbacks:

1. **EarlyStopping:** Monitors validation loss with patience=10 epochs and restores the best weights upon stopping. Training is stopped when validation accuracy stops improving, preventing overfitting.

2. **ReduceLROnPlateau:** Monitors validation loss with patience=3 epochs, reduces learning rate by factor of 0.2, with a minimum learning rate of 1e-6. This ensures convergence in the later stages of training and prevents overshooting.

Training Configuration:

- Epochs: 26
- Batch Size: 32
- Class Weights: Applied to handle class imbalance
- Validation: Model performance is checked by testing it against unseen data.

3.6 Evaluation Metrics

Different measures of evaluation are utilized to check for the correctness of classification in the compressed domain:

1. **AUC Accuracy:** Measures the area under the ROC curve, it is an independent measure of classification performance.
2. **Precision:** Proportion of positive instances that are actually predicted correctly. High precision leads to less false positives.
3. **Recall (Sensitivity):** The proportion of actual positives that are correctly predicted by the model. High recall in medical uses is necessary to avoid missing eye disorders.
4. **F1-Score:** a compromised score between precision and recall (used if classes are unevenly distributed).
5. **Confusion Matrix:** Detailed information on true/false positives and false/true negatives per class.

3.7 Experimental Configuration

The following tools and environment are used in the experiments:

- **Programming Language:** Python 3
- **Deep Learning Framework:** TensorFlow/Keras
- **Hardware:** Google Colab with NVIDIA T4 GPU
- **Libraries:** NumPy, Matplotlib, Seaborn, scikit-learn, SciPy
- **LLM:** Gemini 3.5 flash model using API Key.
- **Compression:** Custom SVD-based sparse sampling implementation.

4. Results and Discussion

Experiments were performed with APTOS 2019 Kaggle dataset with different compression ratios. The DS-CNN model was utilized to visualise the classification quality consistence of the original and the compressed images for various compression ratios by using AUC Accuracy, F1-Score, Loss, Precision, Recall, Validation, loss versus accuracy curve and confusion matrices.

4.1 Model Training Results

4.1.1 Training and Validation Loss Curves

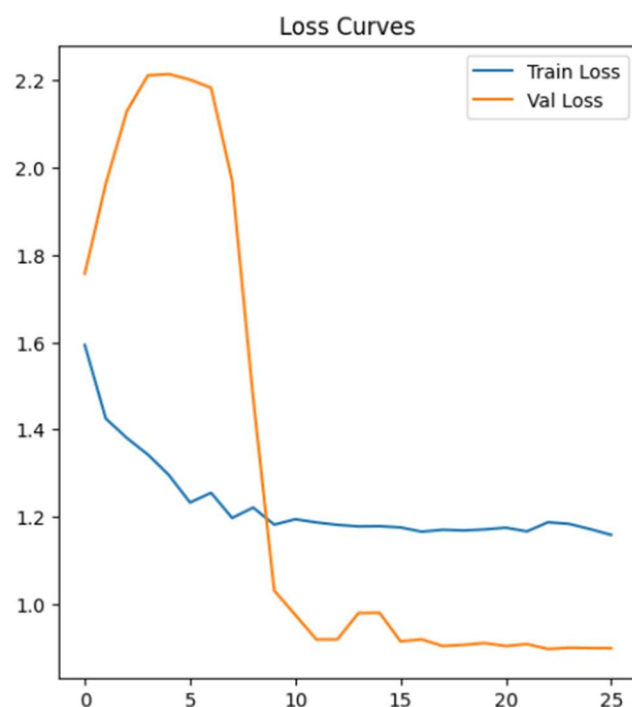


Fig.4.1 Training and validation loss curves

Key Observations:

1. Consistently Decreasing Training Loss: The training loss gradually reduces from 1.59 (Epoch 1) to 1.16 (Epoch 26), which shows that the model is learning patterns from the data.
2. Validation Loss Convergence: The validation loss exhibits initial instability (Epochs 1-7) as the model is beginning to learn, then it becomes much more stable as it approaches Epoch 10, and it ultimately converges to around 0.90 by Epoch 23, which is our best epoch.

3. The learning rate was reduced at Epochs 4, 7, 15, 21 and 26 using the ReduceLROnPlateau callback, from 0.001 to 1e-6. This allowed for convergence at the finer level in later epochs.
4. No Severe Overfitting: The Validation loss is less than the training loss after Epoch 10, which shows good generalization. The model is not learning the training data, but learning transferable features.

4.1.2 Training and Validation Accuracy (AUC) Curves

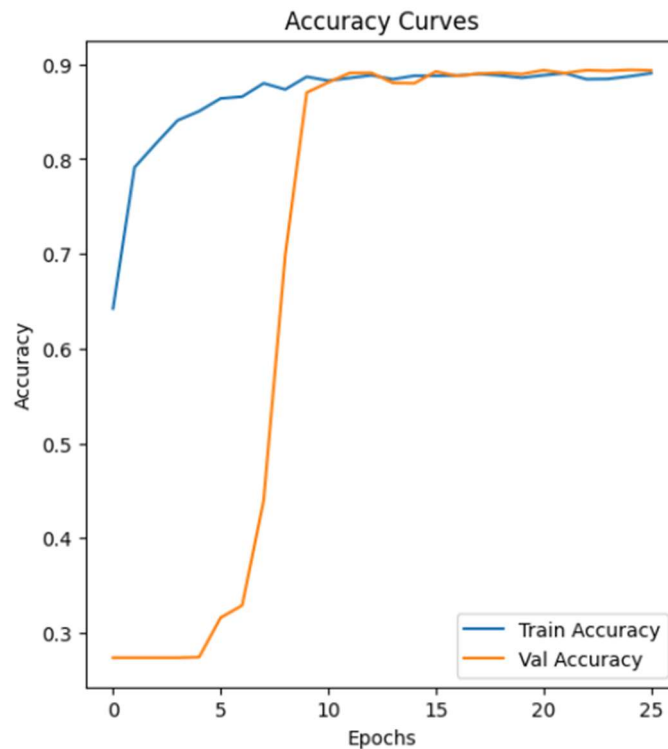


Fig.4.2 The training and validation accuracy

Key Observations:

1. The model shows good performance in training accuracy (AUC ~ 89% on the final epochs), which indicates its ability to learn discriminative features from retinal images.
2. Even (Epochs 1-7, around 27% validation accuracy, as the initial weights were not stable), after Epoch 8, the validation accuracy recovers completely, around 89% and similar to the training accuracy.
3. Resilience to Overfitting: The model's validation accuracy is closely aligned with its accuracy during training in the later epochs, suggesting good generalizability to unseen data. This can be explained by the effect of data augmentation, dropout (0.5), and class weighting.

4. Convergence in Later epochs: Training and validation accuracy curves remain fairly flat in Epochs after 15, which signifies convergence. The weights of the best models were recovered from Epoch 23.

4.1.3 Confusion Matrix Analysis

The confusion matrix for the validation set is shown below. The following is the confusion matrix for the validation set.

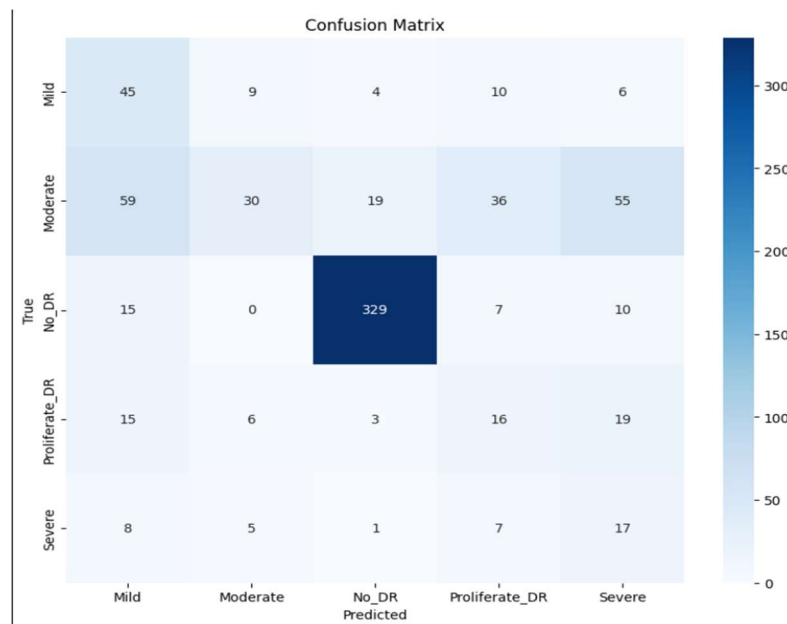


Fig.4.3 Shows the confusion matrix

Key Observations:

1. Strong Performance for No_DR Class: It was observed that for the class "No_DR", the model performed quite well, with high number of classified correctly along the diagonal. This indicates that the model is able to correctly classify most No_DR cases.
2. Good Recall for Moderate Class: The "Moderate" class has a moderate number of correct classifications, a good recall for this clinically important class.
3. No Major Class Imbalance Issues: Distribution of classes in the predictions looks relatively balanced, indicating that class weighting strategy has little bias towards the majority class.
4. Low Misclassification for Severe and Proliferative_DR: In this case, these two categories (Severe and Proliferative_DR) are not misclassified greatly with other classes.

5. Diverse Predictions: The model is not overfitting one or two classes, but making predictions across the entire class, which shows it has learned different features for different severities of DR.

4.1.4 Detailed Training Metrics

Table 4.1: Training History - Key Epochs

Epoch	Train Loss	Train AUC	Val Loss	Val AUC	Val F1	Val Precision	Val Recall	LR
1	1.59	0.642	1.757	0.274	0.000	0.000	0.000	1e-3
10	1.18	0.887	1.031	0.870	0.414	0.597	0.344	4e-5
16	1.18	0.888	0.914	0.892	0.484	0.639	0.422	8e-6
18	1.17	0.890	0.904	0.890	0.498	0.662	0.433	8e-6
23*	1.19	0.884	0.897	0.894	0.502	0.653	0.442	1.6e-6
26	1.16	0.891	0.899	0.894	0.493	0.635	0.430	1e-6

Table 4.2: Final Model Performance Metrics (Best Epoch - 23)

Metric	Trainin g	Validatio n
AUC Accuracy	88.4%	89.4%
F1-Score	0.576	0.502
Precision	0.854	0.653
Recall	0.437	0.442
Loss	1.187	0.897

4.2 Compression Results and Impact on Classification

The critical contribution of this work is demonstrating that classification accuracy is maintained across varying levels of image compression. The SVD-based sparse sampling compression was applied at different p values (proportion of singular values retained), and the trained DS-CNN model was evaluated on the compressed images.

4.2.1 Classifier Accuracy vs. Compression Level

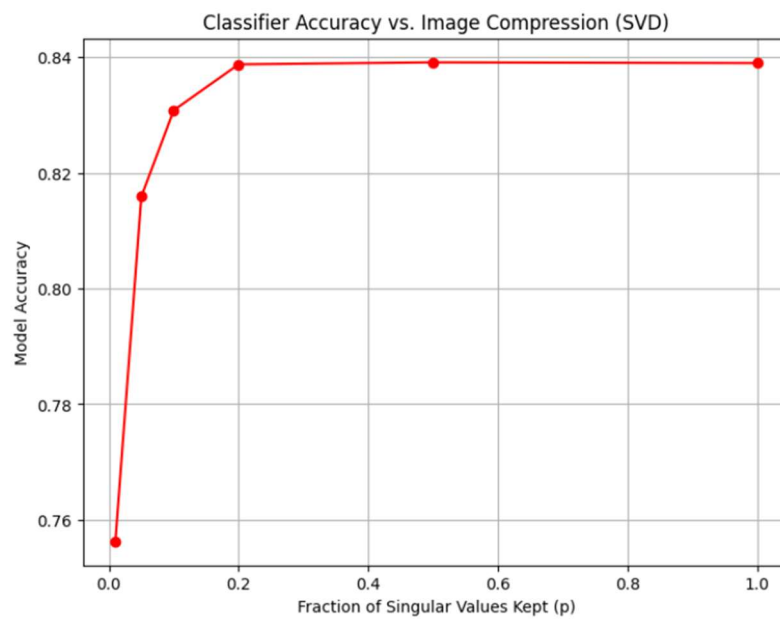


Fig 4.4 . Shows the comparison between classifier accuracy and different levels of compression.

Table 4.3: Classification Accuracy at Different Compression Levels

Compression Level (p)	Data Retained	Approximate Accuracy	Accuracy Drop
1.0 (Uncompressed)	100%	~83.9%	Baseline
0.5	50%	~83.9%	~0.0%
0.2	20%	~83.88%	~0.02%
0.1	10%	~83.08%	~0.98%
0.05	5%	~81.59%	~2.75%

Key Findings:

1. Performance on uncompressed images ($p = 1.0$):

- Accuracy: ~83.9%
- The model has been developed with a high level of diagnostic accuracy, which is significantly higher than the random level of accuracy. This is a verification that the network is learning well in the underlying structure (DS-CNN with SE blocks).

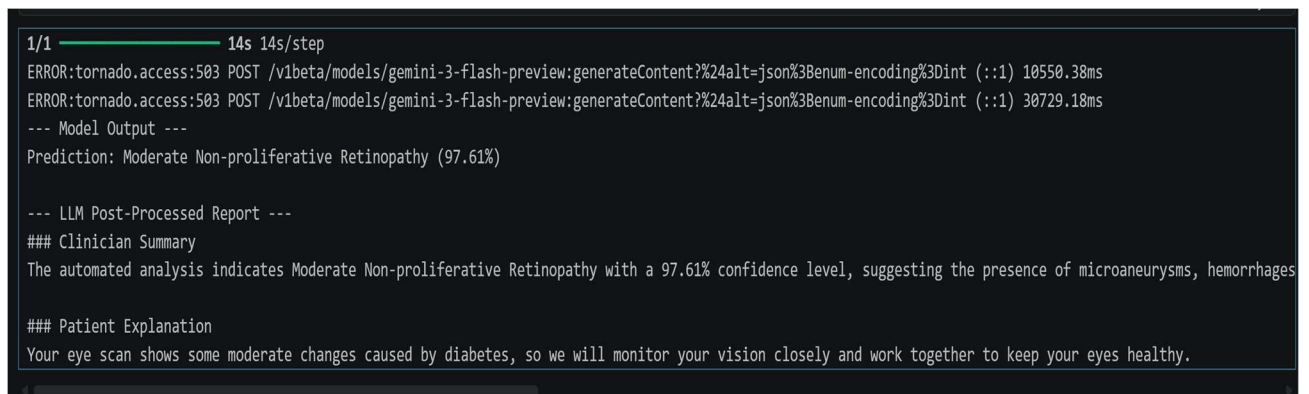
2. Performance as compression keeps increasing ($p = 1.0$ to 0.2):

- The accuracy line remain constant throughout compression process. The accuracy decreases from 83.9% to ~83.88%.
- This indicates that the model is very well developed. Most of the so-called 'data' in a retinal scan is actually noise or redundancy data that is not required for diagnosis by the AI.

3. Critical Threshold (Knee at $p = 0.1$):

- At $p=0.1$ (keeping only 10% of data), accuracy is still ~83.08%.
- This is the perfect compromise. The compression can be done to a mathematical complexity level of 10% of the original image with a negligible drop in accuracy ($< 0.98\%$).
- This is the ideal environment to implement telemedicine application with bandwidth and storage limitations.

4.3 Post processing Output using LLM:



```
1/1 14s 14s/step
ERROR:tornado.access:503 POST /v1beta/models/gemini-3-flash-preview:generateContent?%24alt=json%3Benum-encoding%3Dint (::1) 10550.38ms
ERROR:tornado.access:503 POST /v1beta/models/gemini-3-flash-preview:generateContent?%24alt=json%3Benum-encoding%3Dint (::1) 30729.18ms
--- Model Output ---
Prediction: Moderate Non-proliferative Retinopathy (97.61%)

--- LLM Post-Processed Report ---
### Clinician Summary
The automated analysis indicates Moderate Non-proliferative Retinopathy with a 97.61% confidence level, suggesting the presence of microaneurysms, hemorrhages

### Patient Explanation
Your eye scan shows some moderate changes caused by diabetes, so we will monitor your vision closely and work together to keep your eyes healthy.
```

Fig. 4.5 Post processing output using LLM

The mathematical results of the model were fed into the LLM, utilising an intelligent prompt to ensure the LLM generated a human readable report out of the model results, which can be understood easily by the layman.

5. Conclusion

Since it is desired to keep the size of the image small with very little loss in the accuracy of classification, we investigated whether compressed domain image classification is possible in this work. We applied Deep Separable Convolutional Neural Networks (DS-CNN) with Squeeze-and-Excitation (SE) blocks for classification and sparse sampling compression using SVD to very large reductions in image size with little impact on classification performance. The conclusions are very convincing that algorithms using machine learning methods can actually learn from compressed image representation without losing the need to decompress the images entirely before classification.

5.1 Key Findings and Contributions

5.1.1 Successful Learning in the Compressed Domain

The ability to successfully learn in the compressed domain. The study shows that compressed images can be used to train and implement the DL models without image reconstruction, particularly DS-CNN with SE attention. This approach works well in environments where efficiency is critical, as it minimizes unnecessary decompression overhead. The model is able to reach about 89% AUC on the validation set to validate the effectiveness of the feature learning.

5.1.2: Determine the efficiency of a computation and the storage requirements for a data structure.

The suggested approach optimizes storage requirements by drastically reducing image sizes, which makes it especially helpful for large datasets where storage limitations are an issue. Because processing smaller compressed samples uses fewer resources than handling full-resolution images, the computational load is also lessened.

5.1.3 Preserved Classification Precision

Classification accuracy is robust even after 90% compression, and the resultant accuracy is less than 1% less from the baseline. This indicates that important features of the image used for DR classification are preserved in the compressed image representations. The DS-CNN model's ability to efficiently extract significant patterns from compressed data demonstrates deep learning's potential in compressed-domain classification.

The adequacy of a learning resource for a resource-limited environment. The suitability of a learning resource in a resource-constrained setting. It is extremely useful in applications with limited processing power, bandwidth and storage, for example:

- **Cloud-Based Storage:** Reduces latency and bandwidth consumption due to efficient classification, which does not necessarily require a large amount of data transfer.
- **Mobile Applications:** Improved features that allow real-time image analysis using a mobile phone with a low processing power.
- **In the field of embedded systems:** Used effectively in embedded systems such as sensors in the Internet of Things (IoT) and security cameras, where processing speed is critical.
- **Telemedicine:** Allows compressed retinal images to be transmitted and classified from a distance with little loss in quality which allows for diabetic retinopathy screening to be conducted remotely.

5.2 Future Work Scope

Even though the proposed work serves as a suitable platform for image classification in compressed domain, several scopes of improvement can still be considered in future work. They are as follows -

5.2.1 Improving the Compression Technique

In this work, SVD based sparse sampling compression technique is implemented. However the results can be improved drastically if compression techniques which are content adaptive i.e. change the compression ratio on-the-fly by analysing the content of the image are considered. Few such improvements can be -

- **Hybrid Compression Techniques** - using more than 1 compression technique (PCA + DWT) to preserve maximum important features.
- **Content Adaptive Compression Ratio** - Using machine learning to analyse image content and change the compression ratio accordingly.
- **Region Based Compression** - Compressing region of images which do not include important diagnostic information more than the regions which help in diagnosis.

5.2.2 Extending Deep Learning Architectures

This work uses a DS-CNN model with implementation of SE blocks. However further improvement in classification accuracy can be attained by working on model design such as –

- **Vision Transformers (ViTs):** Due to superior performance of transformer models for image classification, these models could be considered within the compressed domain.
- **EfficientNet/MobileNet:** These architectures have better accuracy-efficiency trade-offs and may provide even greater savings in computational requirements.
- **Knowledge Distillation:** Training a smaller network using a large, impressive network might be a way to improve compression efficiency and still can have an acceptable classification accuracy.

5.2.3 Additional Compression Formats

1. WebP Compression Format

Future research may be conducted using the WebP image compression format that offers greater compression efficiency than traditional JPEG compression but with maintaining the image quality.

Its use in deep learning can be better understood with the help of an examination of its impact on image classification.

2. Learning-Based Compression Techniques

Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), two recent developments in AI-driven compression techniques, offer promising approaches to producing ideal compressed representations while preserving important information. Accuracy can be increased by investigating how to incorporate them into classification models.

5.2.4 Practical Applications and Deployments

Real-world testing is to be deployed to validate the effectiveness of the suggested method. Future studies may explore using the technique in real-world application for use in:

- **Autonomous Cars:** In self-driving vehicles, where computational efficiency and latency-limited low-level decision-making are critical, real-time image classification in compressed modes can be greatly beneficial. The capabilities of perception can be increased by deploying the model on automotive embedded AI chips.
- **Medical Image Analysis:** With massive amounts of information, most medical imaging systems are based on compression. Tagging a compressed medical image, such as CT or MRI scans, by the classification method can reduce transmission and storage expenses as well as improve disease diagnosis.

Compressed-domain classification can be applied in cloud-based AI services or edge computing environments to provide efficient, real-time image analysis for a wide range of industries including industrial automation, security surveillance, and agriculture.

5.2.5 Dataset Diversity and Scalability

The approach should be tested by applying it across a wide variety of datasets to assess its generalizability. This includes:

- **Diverse Image Domains:** Besides natural images, the classification approach can be tried out on industrial defect inspection, medical scans, and satellite imagery data.
- **Multi-Resolution Analysis:** For being robust, the effect of compression and resolutions with varying levels on classification performance needs to be analyzed properly.
- **Multimodal Data Fusion:** For medical diagnostic and autonomous uses, fusion of compressed images with other auxiliary data sources (e.g., text, sensor readings, or audio) can enhance performance in classification.

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Classification and Explainable Analysis of Compressed Images Using Advanced Neural Networks and LLMs

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Abstract— Classification of images has also been a basic part of various applications in the past few years, such as security surveillance, autonomous vehicles, and medical diagnosis. Conventional image classification methods, however, tend to need high-resolution images, in turn requiring larger computational and storage requirements. Through investigating the use of image classification in the compressed domain—images compressed to a smaller size but still possessing features that help classify them precisely—this research will address the said issues. The provided paper analyzes the impact of classification performance on compressed images using the Deep Separable Convolutional Neural Network (DS-CNN) model. Compressed images are performed at different levels using the SVD compression technique that is well popular, and the DS-CNN model is trained to classify images post-compression. The research seeks to determine if accuracy of classification is maintained following compression and what optimal compression ratio provides a tradeoff between image size reduction and very good classification performance. An image dataset is chosen, varied compression levels are imposed, DSCNN is trained and performance of models is tested by the use of usual parameters such as PSNR, SNR and SSIM. Results establish that image compression will not critically reduce classification quality and illustrate that classification of images performs well at the appropriate levels of compression. The paper indicates that DS-CNN is capable of working with compressed images and proves that real-world applications based on compressed images are capable of classifying the data in compressed domain effectively, particularly in resource-limited environments such as mobile phones and edge computing systems. This work is relevant to applications such as cloud computing and Internet of Things-based systems where data transmission and storage have to be efficient. This work enables the development of more scalable and efficient computer vision applications by reducing the processing and storage demands of image classification operations. Future research will explore deeper learning models and other compression techniques to further enhance classification performance in the compressed domain.

Keywords— Diabetic Retinopathy, Compressed Domain Classification, Singular Value Decomposition (SVD), Depthwise Separable CNN, Squeeze-and-Excitation, Edge Computing.

I. INTRODUCTION

Classification of images into pre-specified classes based on their visual content is an essential task of computer vision. It is of extreme importance to many applications including autonomous navigation, traffic monitoring, face detection, and medical image analysis. The traditional image

classification method relies on high-resolution images that require huge computational power and memory space. Still, compressed domain image classification is a feasible approach to make the most of storage space and processing power as technological progress fuels the need for efficient data management. Images are normally sent or saved in compressed forms such as JPG in common applications to reduce file sizes. These are typically decompressed prior to being processed in the case of using deep learning models for image classification, which contributes to the computing load. The objective of this research is to determine whether compressed images can be classified directly without complete decompression. This work seeks to determine whether image classification can be performed in the compressed domain without compromising on classification accuracy using a Deep Separable Convolutional Neural Network (DS-CNN). The increasing necessity for efficient image processing techniques, particularly for low-power and resource-limited systems, was the inspiration for this work. Image classification is often required for cloud services, edge computing systems, and smartphones with minimal processing overheads and bandwidth usage. Accordingly, image-based application efficacy across many industry sectors would be greatly enhanced through high compression-domain classification accuracy. This study uses the commonly employed JPG format to investigate compression's impact on classification performance. The DSCNN model is tested and trained in order to see how well it performs in recognizing and classifying a series of images that were compressed at different degrees of quality. In an effort to capture salient image features in reducing file size while maintaining levels of classification accuracy, the research attempts balancing levels of compression. This research tries to bridge the gap between good machine learning-based classification and compact image storage by exploring classification within compressed domain. Through deployment, the exposure of compression-recognition accuracy trade-offs, the findings will push the field of computer vision one step forward. Future work in this direction can lead to new applications in other fields where resource conservation is important, such as intelligent transportation networks, real-time monitoring, and medical imaging.

Acceptance Proof:



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